

- RAG Learning

3.1 Ratio & Proportion

AQA GCSE Mathematics 8300 — Ratio, Proportion & Rates of Change — spec refs R2–R8, R10, R12, R13 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.1.1 Ratio Notation and Simplifying

A **ratio** compares the sizes of two or more quantities. The ratio $a : b$ means “for every a parts of the first, there are b parts of the second”. Order matters: boys : girls = $3 : 5$ is not the same as girls : boys.

To **simplify** a ratio, divide every part by the highest common factor (HCF). A ratio is in its **simplest form** when the parts are whole numbers with no common factor other than 1.

- Make the **units the same** first, e.g. $40 \text{ cm} : 2 \text{ m} = 40 : 200 = 1 : 5$.
- Clear **decimals** by multiplying by 10, 100, ..., e.g. $1.5 : 4.5 = 15 : 45 = 1 : 3$.
- Clear **fractions** by multiplying by a common denominator, e.g. $\frac{1}{2} : \frac{2}{3} = 3 : 4$ (multiply by 6).

Worked Example: Simplify $2.5 \text{ kg} : 750 \text{ g}$.

Same units: $2500 : 750$. HCF is 250, so $2500 : 750 = 10 : 3$.

Worked Example: Simplify $45 : 60 : 75$.

HCF of all three is 15: $45 : 60 : 75 = 3 : 4 : 5$.

The form $1 : n$ and $n : 1$

Divide both parts by the number you want to become 1. The answer need not be a whole number. This is useful for comparing scales and concentrations.

Worked Example: Write $3 : 8$ in the form $1 : n$.

Divide both parts by 3: $1 : \frac{8}{3} = 1 : 2.67$ (to 2 d.p.).

Exam Tip: "Give your answer in its simplest form" means whole numbers with no common factor. A ratio such as $1 : 2.5$ is *not* simplest form — write $2 : 5$.

3.1.2 Ratios, Fractions and Multiplicative Relationships

One quantity as a fraction of another

To write A as a fraction of B , write $\frac{A}{B}$ (same units) and simplify. The fraction can be **greater than 1**.

Worked Example: (a) Write 15 as a fraction of 40: $\frac{15}{40} = \frac{3}{8}$.

(b) Write 50 cm as a fraction of 20 cm: $\frac{50}{20} = \frac{5}{2}$ — the first length is $2\frac{1}{2}$ times the second.

Ratio to fractions

If boys : girls = $3 : 5$ then there are $3 + 5 = 8$ parts altogether:

- the fraction of the class who are boys is $\frac{3}{8}$; girls is $\frac{5}{8}$;
- the number of boys is $\frac{3}{5}$ of the number of girls, and the number of girls is $\frac{5}{3}$ of the number of boys.

Part : part and part : whole

If $A : B = a : b$, then A is $\frac{a}{a+b}$ of the whole, and $\frac{A}{B} = \frac{a}{b}$, i.e. $A = \frac{a}{b}B$.

Ratios and linear functions

If $x : y = 3 : 4$ then $\frac{y}{x} = \frac{4}{3}$, so $y = \frac{4}{3}x$. Every pair of values in a fixed ratio lies on a **straight line through the origin**; the gradient is the multiplier between the quantities. This links ratio to direct proportion (section 3.1.5).

Exam Tip: A ratio $a : b$ is *not* the fraction $\frac{a}{b}$ of the whole. With $3 : 5$, the first share is $\frac{3}{8}$ of the total, not $\frac{3}{5}$.

3.1.3 Dividing in a Ratio and Ratio Problems

Sharing in a ratio $a : b : c$

1. Total parts = $a + b + c$
2. One part = amount $\div$ total parts
3. Each share = its ratio number $\times$ one part

Worked Example: Share £450 in the ratio 4 : 5.

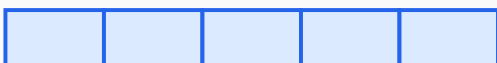
$4 + 5 = 9$ parts. One part = $450 \div 9 = 50$.

Shares: $4 \times 50 = \text{£}200$ and $5 \times 50 = \text{£}250$. Check: $200 + 250 = 450$. ✓

When you know one share or a difference

If you are given one share, or the difference between shares, find the value of **one part** from that information — do not divide the total.

Boys



Girls



2 parts = 14 more boys

Worked Example: Boys : girls in a club = $5 : 3$. There are 14 more boys than girls. How many members are there?

The difference is $5 - 3 = 2$ parts, so 2 parts = 14 and 1 part = 7.

Total = 8 parts = $8 \times 7 = 56$ members.

Combining ratios

Worked Example: $A : B = 3 : 4$ and $B : C = 6 : 5$. Find $A : B : C$.

Make the B values equal: LCM of 4 and 6 is 12.

$A : B = 9 : 12$ and $B : C = 12 : 10$, so $A : B : C = 9 : 12 : 10$.

Ratio problems with algebra

When a ratio *changes* (counters added or removed, ages increasing), call the parts $3k$, $7k$, ... and form an equation.

Worked Example: Red : blue counters = $3 : 7$. Kim removes 12 blue counters and the ratio becomes $1 : 2$. How many counters were there at first?

Red = $3k$, blue = $7k$. After: $3k : (7k - 12) = 1 : 2$, so $2 \times 3k = 7k - 12$.

$6k = 7k - 12 \Rightarrow k = 12$. Original total = $10k = 120$.

Real contexts: mixing, recipes, conversion

For limiting-ingredient problems, work out how much of the product each ingredient allows, then take the **smallest**.

Worked Example: Concrete: cement : sand : gravel = $1 : 2 : 4$ (7 parts). A builder has 50 kg cement, 120 kg sand, 160 kg gravel.

Cement allows 1 part = 50 kg $\Rightarrow$ 350 kg concrete. Sand: 1 part = 60 kg $\Rightarrow$ 420 kg. Gravel:

1 part = 40 kg $\Rightarrow$ 280 kg.

Greatest amount = **280** kg (gravel runs out first).

Exam Tip: Always check your shares add up to the total, and re-read which share the question asks for (smaller, larger, or the difference).

3.1.4 Unitary Method, Best Buys, Scales and Maps

The unitary method and best buys

Find the value of **one** item (or one gram, one litre, ...) and then multiply. For best buys, compare the **cost per unit** (lower is better) or the **amount per £1** (higher is better).

Worked Example: Juice: 1 litre for £1.35, 1.5 litres for £1.95, 2 litres for £2.70. Which is best value?

Cost per litre: £1.35, $1.95 \div 1.5 = \pounds 1.30$, $2.70 \div 2 = \pounds 1.35$.

The **1.5 litre** carton is best value.

Worked Example: Exchange rate £1 = €1.15. Convert €506 to pounds.

Euros $\rightarrow$ pounds: divide. $506 \div 1.15 = \pounds 440$.

Scale factors, scale drawings and maps

A map scale $1 : n$ means 1 unit on the map represents n of the **same unit** in real life. Real = map $\times n$; map = real $\div n$.

Scale	1 cm on the map represents
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1 : 100	100 cm = 1 m
1 : 25 000	25 000 cm = 250 m
1 : 50 000	50 000 cm = 500 m
1 : 100 000	100 000 cm = 1 km

Worked Example: Scale 1 : 50 000. Two villages are 7.5 km apart. How far apart are they on the map?

7.5 km = 750 000 cm. Map distance = $750\,000 \div 50\,000 = 15$ cm.

Worked Example: A field is drawn at a scale of 1 : 2000 as a 6 cm by 4.5 cm rectangle. Find its real area.

Real lengths: $6 \times 2000 = 12\,000$ cm = 120 m and $4.5 \times 2000 = 9000$ cm = 90 m.

Area = $120 \times 90 = 10\,800$ m². (Convert the lengths first — do not multiply the area by 2000.)

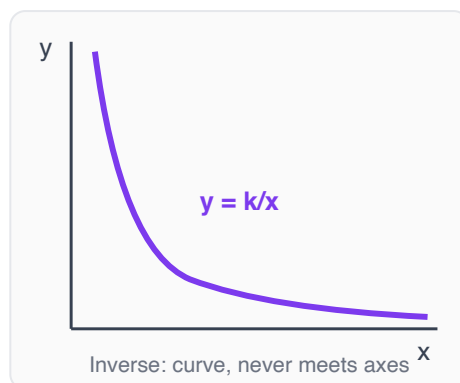
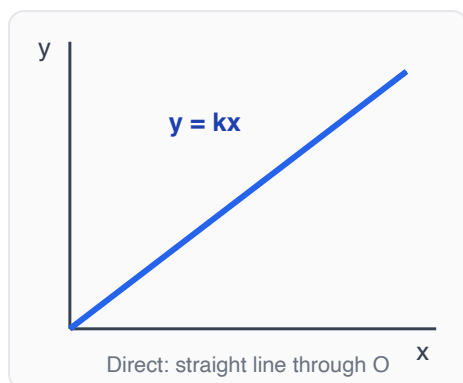
Exam Tip: Map and scale questions often combine with bearings and constructions. Keep units consistent: 1 m = 100 cm, 1 km = 100 000 cm.

3.1.5 Direct and Inverse Proportion

Two ratios are **in proportion** when they are equal, e.g. $6 : 15 = 10 : 25$ (both = $2 : 5$).

y is **directly proportional** to x , written $y \propto x$, if $y = kx$ for a constant k : when x doubles, y doubles.

y is **inversely proportional** to x , written $y \propto \frac{1}{x}$, if $y = \frac{k}{x}$, i.e. $xy = k$: when x doubles, y halves.



Solving proportion problems numerically

Worked Example: Direct: $y \propto x$ and $y = 18$ when $x = 4$. Find y when $x = 10$.

$k = 18 \div 4 = 4.5$, so $y = 4.5x$. When $x = 10$: $y = \mathbf{45}$.

Worked Example: Inverse: 6 workers harvest a field in 15 hours. How long would 9 workers take?

Total work = $6 \times 15 = 90$ worker-hours (constant). Time = $90 \div 9 = \mathbf{10 \text{ hours}}$.

To finish in 5 hours you would need $90 \div 5 = 18$ workers. (Assumes all work at the same rate.)

	Direct proportion	Inverse proportion
Statement	$y \propto x$	$y \propto \frac{1}{x}$
Equation	$y = kx$	$y = \frac{k}{x}$
What stays constant	$\frac{y}{x} = k$	$xy = k$
x doubles	y doubles	y halves
Graph	straight line through the origin	reciprocal curve

Exam Tip: Ask "if one quantity goes up, does the other go up or down?" More workers $\Rightarrow$ less time, so that is inverse proportion.

3.1.6 Proportion Equations Higher

At Higher tier you must **construct** the equation for a proportion relationship involving powers and roots, find k , and use it.

Method

1. Write the relationship with k , e.g. $y = kx^2$, $y = \frac{k}{\sqrt{x}}$, $F = \frac{k}{d^2}$
2. Substitute the given pair of values and solve for k
3. Rewrite the formula using the value of k
4. Substitute to find the required value

In words	Equation	If x is doubled, y is...
y is proportional to the square of x	$y = kx^2$	multiplied by 4
y is proportional to the cube of x	$y = kx^3$	multiplied by 8
y is proportional to the square root of x	$y = k\sqrt{x}$	multiplied by $\sqrt{2}$
y is inversely proportional to the square of x	$y = \frac{k}{x^2}$	divided by 4
y is inversely proportional to the square root of x	$y = \frac{k}{\sqrt{x}}$	divided by $\sqrt{2}$

Worked Example: y is directly proportional to x^2 . When $x = 4$, $y = 48$. Find y when $x = 2.5$.

$$y = kx^2 \Rightarrow 48 = 16k \Rightarrow k = 3, \text{ so } y = 3x^2.$$

$$y = 3 \times 2.5^2 = 3 \times 6.25 = \mathbf{18.75}.$$

Worked Example: The force F between two magnets is inversely proportional to the square of the distance d . $F = 20$ when $d = 3$. Find d when $F = 5$.

$$F = \frac{k}{d^2} \Rightarrow 20 = \frac{k}{9} \Rightarrow k = 180, \text{ so } F = \frac{180}{d^2}.$$

$$5 = \frac{180}{d^2} \Rightarrow d^2 = 36 \Rightarrow d = \mathbf{6}.$$

Worked Example: T is inversely proportional to $\sqrt{L}$. $T = 6$ when $L = 16$. Find T when $L = 36$.

$$T = \frac{k}{\sqrt{L}} \Rightarrow 6 = \frac{k}{4} \Rightarrow k = 24. \text{ When } L = 36: T = \frac{24}{6} = \mathbf{4}.$$

Percentage changes in proportion problems

If $y \propto x^3$ and x increases by 20%, then y is multiplied by $1.2^3 = 1.728$, an increase of **72.8%**.

Exam Tip: Always write the formula with the value of k as a separate line (" $y = 3x^2$ ") — it usually earns its own mark, and it is the answer if the question asks you to "find a formula".

3.1.7 Comparing Lengths, Areas and Volumes

Ratio notation can compare lengths, areas and volumes. For simple shapes you can calculate each area or volume and write the ratio. Enlarged (similar) shapes have corresponding lengths in the same ratio, which links ratio to **similarity** and to the **trigonometric ratios** (in similar right-angled triangles, ratios such as $\frac{\text{opp}}{\text{hyp}}$ are equal for the same angle).

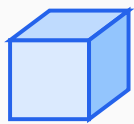
Worked Example: Rectangle A is 3 cm by 4 cm and rectangle B is 6 cm by 8 cm.

Lengths ratio = $3 : 6 = 1 : 2$. Areas: $12 : 48 = 1 : 4$. The area ratio is the square of the length ratio.

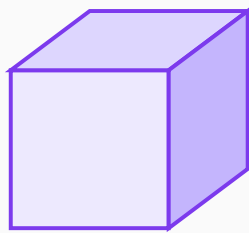
Similar shapes

Higher

If the length scale factor is k (length ratio $a : b$):
 area scale factor = k^2 (area ratio $a^2 : b^2$)
 volume scale factor = k^3 (volume ratio $a^3 : b^3$)



side 1



side 2

L 1 : 2
 A 1 : 4
 V 1 : 8

Worked Example: Two similar bottles are 12 cm and 18 cm tall. The smaller holds 400 ml and has surface area 270 cm^2 .

$$k = 18 \div 12 = 1.5.$$

$$\text{Capacity of larger} = 400 \times 1.5^3 = 400 \times 3.375 = \mathbf{1350 \text{ ml}}.$$

$$\text{Surface area of larger} = 270 \times 1.5^2 = 270 \times 2.25 = \mathbf{607.5 \text{ cm}^2}.$$

Worked Example: Two similar cones have volumes 54 cm^3 and 250 cm^3 . Find the ratio of their heights.

$$\text{Volume ratio } 54 : 250 = 27 : 125. \text{ Length ratio} = \sqrt[3]{27} : \sqrt[3]{125} = \mathbf{3 : 5}.$$

Exam Tip: Going from area or volume back to length, take the square root or cube root of the scale factor *first*. Never multiply a volume by the length scale factor.

Summary

Skill	Key idea
Simplify a ratio	Same units, then divide by the HCF
Share in a ratio	Total parts → one part → each share
Fraction of a whole	$a : b$ gives $\frac{a}{a+b}$ and $\frac{b}{a+b}$
Best buy	Compare cost per unit
Map scale $1 : n$	Real = map $\times n$, then convert units
Direct proportion	$y = kx$; straight line through the origin
Inverse proportion	$y = \frac{k}{x}$; xy constant
Proportion equations Higher	Write with k , substitute, find k , use formula
Similar shapes Higher	Lengths k , areas k^2 , volumes k^3